

MATH 2211, SPRING 2026: SOLUTIONS TO HOMEWORK 7

- (1) Suppose that $A \in M_{m \times n}(F)$. For each of the following, prove the assertion or find a counterexample:
 (a) $\text{rank}(AB) \leq \text{rank}(A)$ for any $n \times p$ -matrix B (p arbitrary);

Solution. We can think of A as coming from a linear transformation $T : F^n \rightarrow F^m$ and B as coming from a linear transformation $S : F^p \rightarrow F^n$. The matrix AB corresponds to the composition $T \circ S : F^p \rightarrow F^m$. Note that everything in the image of $T \circ S$ is already in the image of T (why?). Therefore, we have $\text{im } AB \subset \text{im } A$. Since the rank is the dimension of the image, we see that we must always have $\text{rank}(AB) \leq \text{rank}(A)$.

- (b) $\text{rank}(BA) \leq \text{rank}(A)$ for any $p \times m$ -matrix B (p arbitrary).

Solution. We can think of A as coming from a linear transformation $T : F^n \rightarrow F^m$ and B as coming from a linear transformation $S : F^m \rightarrow F^p$. The matrix BA corresponds to the composition $S \circ T : F^n \rightarrow F^p$. Now, the image of $S \circ T$ is that of the restriction of S to the subspace $\text{im } T \subset F^m$, which is a linear transformation $\text{im } T \rightarrow F^p$. As such, its image cannot have dimension greater than that of its codomain. This tells us exactly that $\text{rank}(BA) \leq \text{rank}(A)$.

Hint: It might be useful to convert to questions about images of linear transformations.

- (2) If $A \in M_{n \times n}(F)$ is an invertible matrix with inverse A^{-1} show that we have

$$AA^{-1} = I_n = A^{-1}A.$$

Solution. A corresponds to an invertible linear transformation $T : F^n \rightarrow F^n$ and A^{-1} corresponds to its inverse $T^{-1} : F^n \rightarrow F^n$. Therefore, the property we seek is a consequence of the fact that $T \circ T^{-1}$ and $T^{-1} \circ T$ are both the identity linear transformation. See Fact in Lecture 13.

- (3) Show that the associated linear transformation

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$\begin{bmatrix} a \\ b \end{bmatrix} \mapsto \begin{bmatrix} 2a + b \\ a + 2b \end{bmatrix}$$

is diagonalizable.

Solution. This is given by the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. The characteristic polynomial of this matrix is $x^2 - 4x + 3 = (x - 3)(x - 1)$. Therefore, the eigenvalues are 1 and 3.

Now, $A - 1 \cdot I_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ has $v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ in its kernel. This means that $T(v_1) = 1 \cdot v_1 = v_1$.

Similarly, $A - 3 \cdot I_2 = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$ has $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ in its kernel. This means that $T(v_2) = 3v_2$.

This gives us a basis $\mathcal{B} = (v_1, v_2)$ with $[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$, and so shows that T is diagonalizable. To check that this is basis of course, we can simply note that v_2 is not a multiple of v_1 .

- (4) Consider the matrix

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Use the previous problem to compute A^n for any integer n .

Solution. The problem had a typo where A was given as $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$. If you used that matrix instead, then the same process that will follow below works, except that the eigenvalues are -1 and 3 , the basis for diagonalization is the same, and you will get

$$A^n = \frac{1}{2} \begin{bmatrix} 3^n + (-1)^n & 3^n - (-1)^n \\ 3^n - (-1)^n & 3^n + (-1)^n \end{bmatrix}$$

Clearly, $[T]_{\mathcal{B}}^n = \begin{bmatrix} 1 & 0 \\ 0 & 3^n \end{bmatrix}$. As explained in Lecture 23, if $\mathcal{B}' = (e_1, e_2)$ is the standard basis, then we have

$$A = [T]_{\mathcal{B}'} = C[T]_{\mathcal{B}}C^{-1}$$

where $C = [v_1 \ v_2] = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$. It is not difficult to compute the inverse in this case (see Remark in Lecture 23:

$$C^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

We then have

$$A^n = C[T]_{\mathcal{B}}^nC^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3^n \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 3^n + 1 & 3^n - 1 \\ 3^n - 1 & 3^n + 1 \end{bmatrix}$$

Definition 1. The **transpose** of a matrix $A \in M_{m \times n}(F)$ with columns

$$A = [v_1 \ v_2 \ \cdots \ v_n]$$

is the matrix $A^T \in M_{n \times m}(F)$ with rows given by

$$A^T = \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{bmatrix}$$

- (5) Show that the linear transformation $U : F^n \rightarrow F^m$ associated with A is surjective if and only if the linear transformation $U' : F^m \rightarrow F^n$ associated with A^T is injective.

Solution. Saying that U is surjective is equivalent to saying that the image has dimension m . In other words, we are saying that the RREF of A has m leading 1s. But the transpose of this RREF will have no free variables (why?). This implies that the transpose of RREF(A) corresponds to an injective linear transformation (why?). Now, what is the connection between A^T and the transpose of RREF(A)? We get from A to RREF(A) using row operations, and therefore, we get from A^T to RREF(A)^T using column operations. This still does not affect injectivity: A system $[A \mid 0]$ will have non-zero solutions precisely when it continues to have non-zero solutions after column operations on A (why?).